Anisa Shaik

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AI Engineer with 3+ years of software engineering experience, including 1+ year building LLM-powered applications and agentic AI systems. Experienced in RAG, multi-agent orchestration, structured tool calling, LLM evaluation, and production AI systems, with a strong foundation in distributed systems and backend engineering.

EDUCATION

Master of Science in **Computer Science** Aug 2024 – May 2026
***University of Florida**, Gainesville, Florida* CGPA – **3.93/4.00**
Coursework: Artificial Intelligence & Machine Learning, Natural Language Processing, Large Language Models, Intro to Data Science, Cloud computing, Mathematics for Intelligent Systems, Advanced Data Structures, Analysis of Algorithms

EXPERIENCE

University of Florida – ***AI Engineer*** May 2025 – Present
- Architected and deployed scalable AI agent workflows for iChatBio (ichatbio.org) a live A2A multi-agent research platform serving 1000+ users using LangGraph, Langchain, tool calling, Redis, and asynchronous processing.
- Optimized agent routing with intent-based conditional execution and parallel tool calls, reducing median query latency by 35% while eliminating unnecessary agent/tool invocations.
- Enforced Pydantic-validated inputs and Instructor-structured LLM outputs, reducing malformed responses by 80% and improving downstream tool-call reliability.
- Built a 3-layer evaluation pipeline with pytest, DeepEval (LLM-as-judge, 500+ cases), and LangSmith tracing, gating deployments and reducing response-quality debugging time by 50%.

Fidelity National Information Services (FIS) – *Software Engineer* Jun 2022 – Jul 2024
- Developed scalable Java microservices using Spring Boot and Docker for a high-volume EBT transaction platform processing 9.4M+ transactions monthly.
- Designed versioned REST APIs supporting 20+ state integrations, implementing schema validation, backward-compatible contracts, and standardized error handling that reduced API error rates by ~15%.
- Built responsive React and TypeScript modules integrated with REST APIs, implementing reusable UI components, client-side validation, and error handling to improve usability across transaction workflows.
- Implemented OAuth2, JWT, and Spring Security, enforcing RBAC and MFA flows, reducing unauthorized access attempts by 30%.
- Diagnosed production transaction failures using SQL query analysis, Linux log tracing, and Splunk correlation; implemented code-level fixes for data-validation and API-handling issues, reducing high-severity incidents by 25%.
- Automated REST/SOAP regression coverage with JavaScript, Postman, and SQL validation, reducing regression cycles from 5 days to 48 hours.

PROJECTS

HALO-RAG: Self-Verification for Hallucination Reduction in RAG **GitHub**
- Engineered a modular RAG pipeline combining hybrid dense-sparse retrieval (FAISS + BM25), cross-encoder reranking, claim-level NLI verification across a 5,000-query SQuAD v2.0 benchmark.
- Fine-tuned FLAN-T5-Large using 4-bit QLoRA for memory-efficient adaptation, improving factual precision from 90.78% to 96.08%.
- Designed a 14+ metric evaluation framework spanning retrieval, factuality, hallucination, abstention, and response quality; measured hallucination reduction from 9.22% to 0.00% with verification and adaptive revision.

Graph Neural Network Recommender **GitHub**
- Implemented a PinSage-based recommendation engine using PyTorch Geometric, leveraging random-walk neighborhood sampling, importance-weighted pooling, and FAISS for scalable item retrieval across 162K+ users and 62K+ movies.
- Improved Hit Rate@10 by 22% over a mean-pooling GNN baseline using importance pooling and hard negative sampling, evaluated on a leakage-free temporal split.

SKILLS

Languages: Python (NumPy, Pandas), SQL, Java, JavaScript
AI / Generative AI: NLP, Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, Fine-tuning, LoRA/QLoRA, LLM APIs, Tool Calling
Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, LangGraph, LangChain, LlamaIndex, MCP, Claude Agents SDK, FastAPI, DeepEval, RAGAS, TensorRT-LLM
Cloud & MLOps: AWS (SageMaker, Bedrock, EKS, Lambda, Fargate), Docker, Kubernetes, MLflow
Databases & Vector Retrieval: PostgreSQL, MySQL, FAISS
Software Engineering: REST APIs, OAuth 2.0, Microservices, Kafka, CI/CD, Git, Linux, Jira, Agile, Scrum, SDLC
Developer Tools: Claude Code, Codex, GitHub Copilot